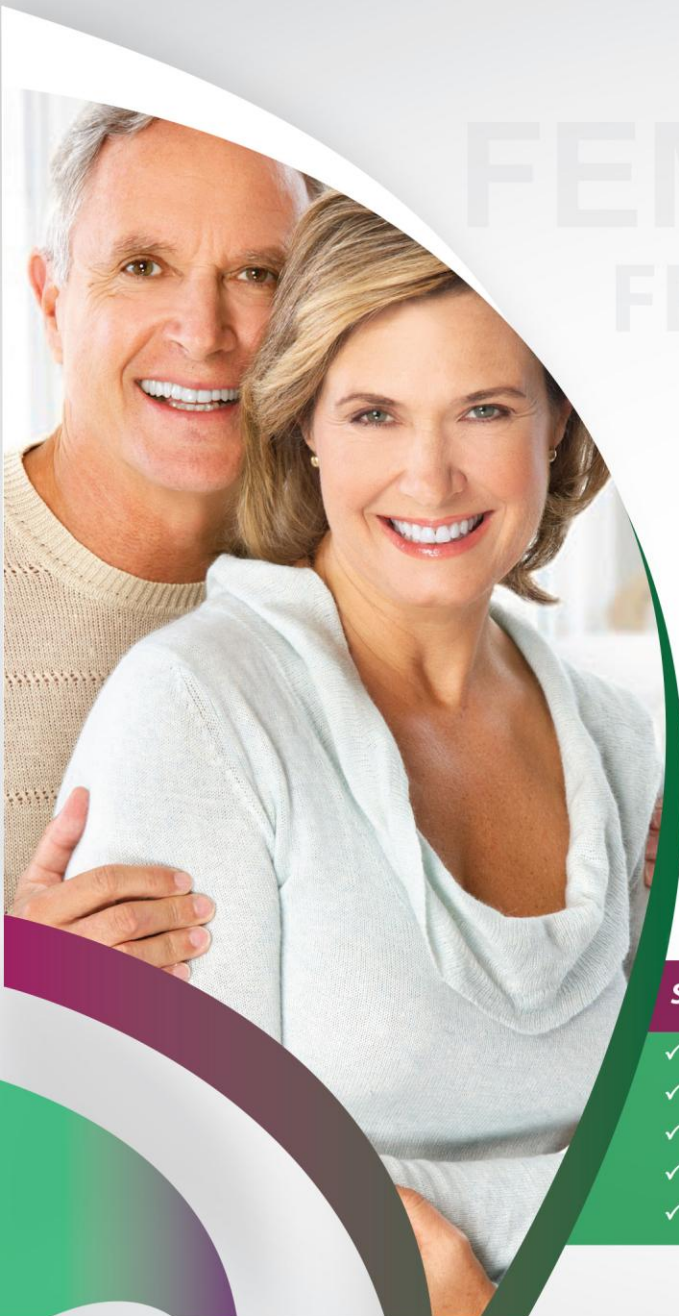




FEMALE
FERTILITY



FibroMed

PHYSICIAN FORMULATED

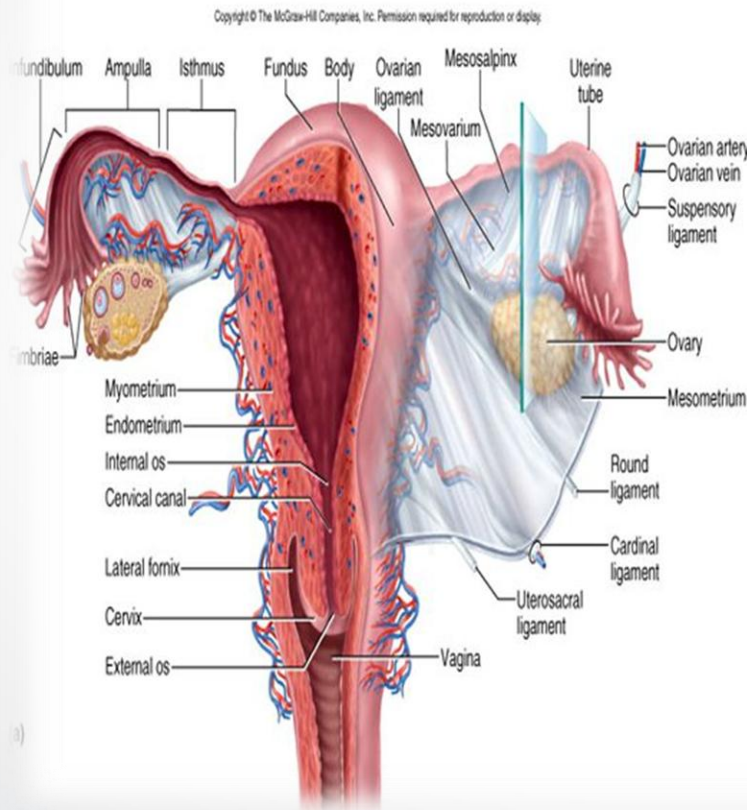


STRONGER FORMULA

- ✓ Goldenseal Root
- ✓ Burdock Root
- ✓ Chaste Berry
- ✓ Red Clover
- ✓ Echinacea

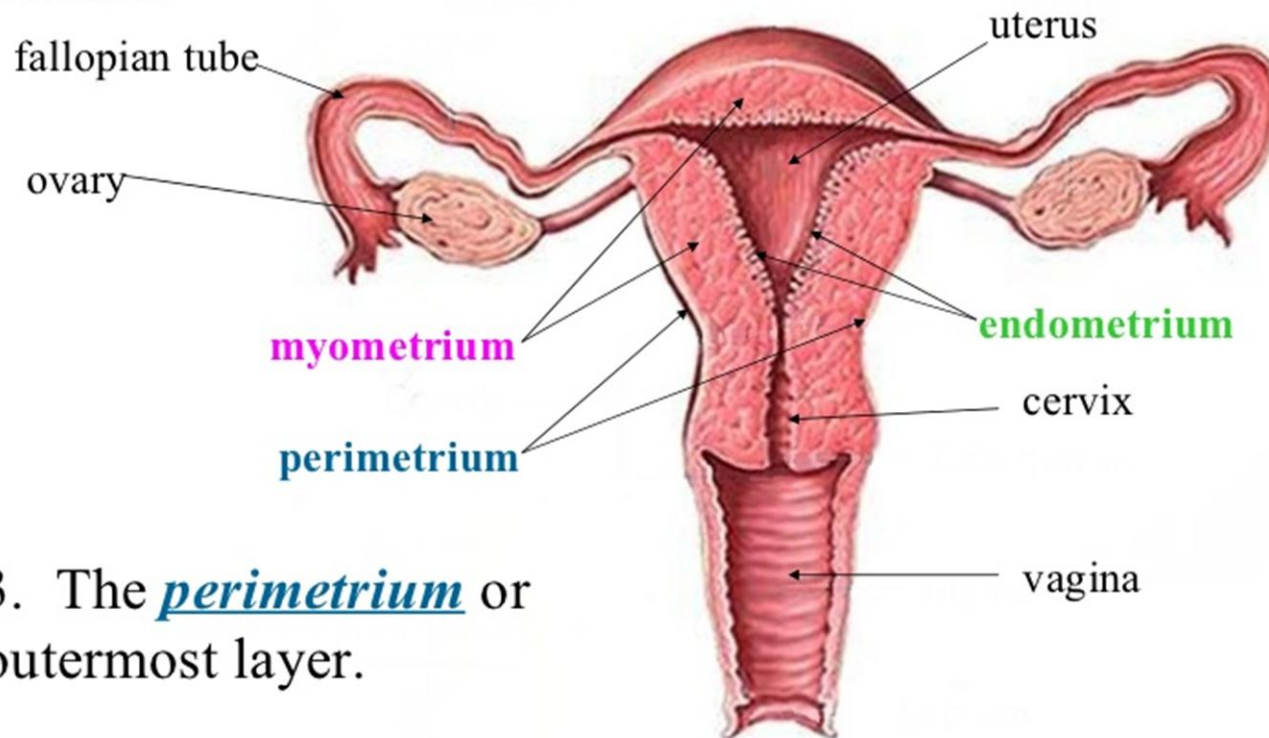


Uterine Wall



- Uterine Wall is composed of three layers
 - Perimetrium – outermost serous layer; the visceral peritoneum
 - Myometrium – middle layer; interlacing layers of smooth muscle
 - Endometrium – mucosal lining of the uterine cavity changes in thickness during the menstrual cycle

2. The myometrium which is the thick muscular, middle layer composed mostly of bundles of *smooth muscle*.



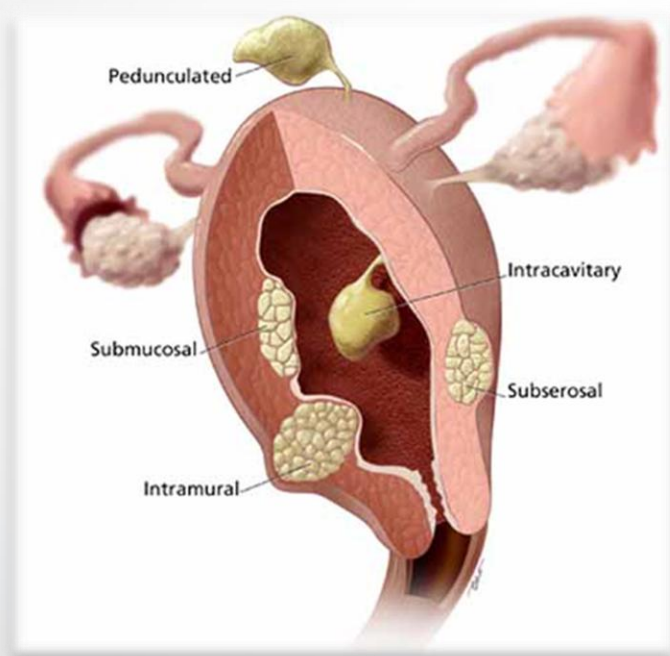
3. The perimetrium or outermost layer.



UTERINE FIBROIDS

- ❖ Uterine leiomyomas (also referred to as fibroids or myomas) are benign tumors arising from the smooth muscle cells of the **myometrium**, known to clinicians as a worldwide public health problem
- ❖ They arise in reproductive-age women
- ❖ The prevalence of fibroids increases with age during the reproductive years
- ❖ Most common pelvic tumor in women
- ❖ 70–80% cumulative incidence in the childbearing years
- ❖ Prevalence is much higher since many fibroids may be asymptomatic

PATHOGENESIS



- ❖ The pathogenesis of fibroids is not well understood
- ❖ Genetic predisposition, environmental factors, steroid hormones, and growth factors important in fibrotic processes and angiogenesis all play a role in the formation and growth of uterine fibroids
- ❖ The disease is heterogeneous and different fibroids may have different etiologies; many may have multifactorial pathogenesis

During a woman's reproductive years her estrogen and progesterone levels are high. When estrogen levels are high, especially during pregnancy, fibroids tend to swell. When estrogen levels are low fibroids may shrink, e.g. during a woman's menopause

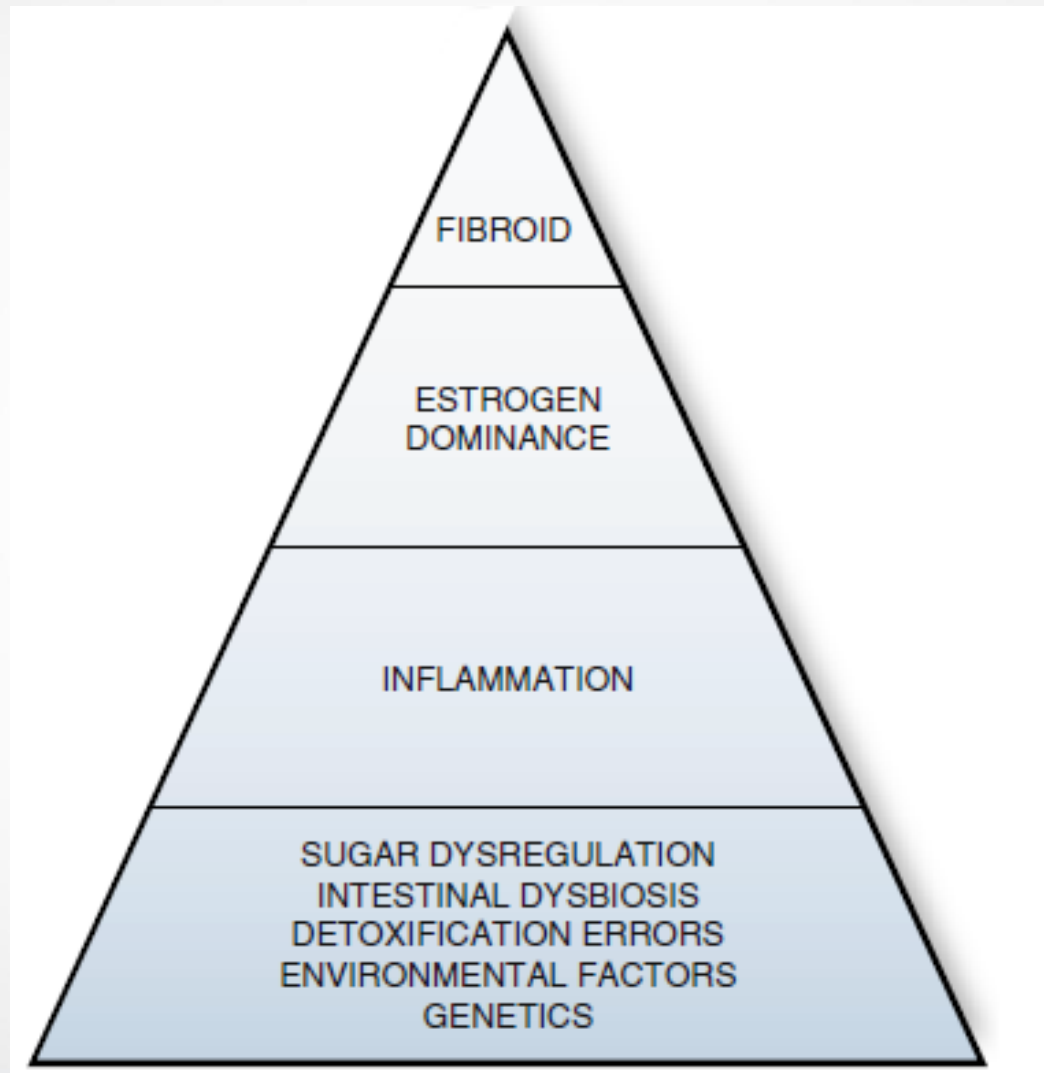
PATHOGENESIS

- The term "**estrogen dominance**" probably sounds familiar. Symptoms include heavy periods, thinning hair, and moody/ aggressive-type PMS.
- One of the most troubling **complications of estrogen dominance** is uterine **fibroids**, the most common type of tumor of the uterus.
- Uterine fibroids develop under the influence of both progesterone and estrogen. Although the conventional medical community remains "unsure" about the role of environmental toxins in endocrine-based health issues, many sources cite plastics, particularly BPA and phthalates, as major influences.

PATHOGENESIS

Larger uterine fibroids are more prevalent among:

- African American women
- Obese women (hormones are fatty substances, and overweight people have higher hormone loads)
- Women who have never had a full-term pregnancy (during pregnancy, the monthly cycle of hormones is suspended temporarily, giving the endocrine system a break)



PATHOGENESIS

Angiogenesis

- Angiogenesis is a multistep cellular process that involves endothelial cell (EC) proliferation, migration, tube formation, and ECM degradation.
- It has been suggested that angiogenesis may play an important role in the regulation of leiomyoma growth.

PATHOGENESIS

Inflammatory mediators

- Considerable evidence suggests that uterine leiomyoma development may be triggered, at least in part, by a chronically active inflammatory immune system.
- The concept of inflammation actually fits into a theory of fibroid development based on an altered response to noxious stimuli; possibly tissue injury from extravasated menstrual blood into the myometrium, or hypoxia leading to altered tissue repair and Fibrosis.
- Among major cytokines, the expression of IL-1, IL-6, IL-11, IL-13, IL-15, IL-33, tumor necrosis factor (TNF)-, granulocyte-macrophage colony-stimulating factor have been implicated with their biological relevance to leiomyoma pathophysiology.

CLINICAL FEATURES



- ❖ The majority of fibroids are small and asymptomatic, but many women with fibroids have significant problems that interfere with some aspect of their lives and warrant therapy
- ❖ **Symptoms are related to the number, size, and location of the tumors.** Fibroids can occur as single or multiple tumors and range in size from microscopic to tens of centimeters
- ❖ Heavy or prolonged menstrual bleeding that may lead to anemia
- ❖ Pelvic pressure or pain
- ❖ Urinary tract or bowel issues, such as difficulty emptying the bladder and constipation
- ❖ Venous compression, Very large uteri may compress the vena cava, leading to an increase in thromboembolic risk
- ❖ Painful menses, Painful intercourse

Summary Slide

Fibroids: Myometrium – middle layer of the uterus

Very common condition – Reproductive age

So many factors or risk factors – family history – toxins- lifestyle – hormonal fluctuations –
ESTROGEN DOMINANCE

Associated with:

Estrogen dominance, Inflammation, Angiogenesis, Detoxification errors

Symptoms:

Pain, heavy bleeding, heavy menstruation, pelvic pressure, difficulties/ issues urination & bowel movement.

That's why we need:

- Hormonal balancing ing
- Anti inflammatory ing
- Anti oxidants ing
- Immunity booster
- Relief of pain
- Detoxification enhancers

MANAGEMENT

- A simple ultrasound is a good way to diagnose uterine fibroids
- If a fibroid is 4 cm in diameter (the size of a small lemon), natural treatments can be very effective
- If the fibroid is between 4 and 8 cm, natural treatments can help, although they take longer to work

MANAGEMENT

- But if the fibroid is larger than 8 cm (grapefruit sized), it generally cannot be treated naturally.
- At this point uterine artery embolization is recommended, which cuts off the blood supply to the fibroid with a minimally invasive surgical procedure.
- **Once shrunk, natural methods can be applied to keep the fibroid small** enough to avoid complications, such as very heavy menstrual bleeding, difficulty with conception, bladder dysfunction (from pressure), and pelvic pain.

MANAGEMENT

Aromatase inhibitor drugs are another common conventional therapy. They work by disrupting a process called aromatization, in which testosterone is converted to estrogen. As most pharmaceuticals, they also create a host of unwanted side effects, including promoting ovarian cysts. They also do nothing to control heavy uterine bleeding.

INFERTILITY OR OBSTETRIC COMPLICATIONS

- ❖ Fibroids that distort the uterine cavity (sub-mucosal or intramural with an intra-cavitary component) result in
 - **Difficulty conceiving a pregnancy**
 - **Increased risk of miscarriage**

- ❖ In addition, fibroids have been associated with adverse pregnancy outcomes such as
 - **Placental abruption**
 - **Fetal growth restriction**
 - **Mal presentation**
 - **Preterm labor and birth**



AMS® FibroMed was especially formulated to

- ✓ Establish hormonal balance
- ✓ Help prevent growth of fibroids
- ✓ Support regulation of menstrual cycle
- ✓ Help suppress over-production of estrogen

AMS® FibroMed is an all natural dietary supplement created to improve various symptoms experienced by women with fibroids (abnormal growths that develop in the uterus), such as pelvic pain/cramps, bloating and weight gain

AMS® FibroMed contains a unique formula with chaste berry and red clover as key ingredients. Chaste berry stops over-production of estrogen, alleviates the severity of PMS, and regulates hormones. Red clover improves circulatory function and regulates the menstrual cycle. In addition to other natural ingredients that have affinity for the female uterus and ovaries (site of cysts and tumors)

The one of a kind formulation of **AMS® FibroMed** maintains the health of ovaries and uterus and helps women with fibroids in a natural manner



Indications

- Small/Medium multiple fibroids
- Pre and Post operation
- Inoperable patients
- Patients who refuse surgery



THE EFFECT OF FIBROMED IS DUE TO

- Motherwort
- Siberian Ginseng
- Goldenseal Root
- Chaste Berry
- Licorice Root
- Burdock Root
- Red Clover
- Echinacea



SUPPLEMENT

Serving Size: 2 Capsules |

Amount Per Serving

Motherwort (*Leonurus Heterophyll*)

Eleuthero Root

(*Eleutherococcus Senticosus*)(Root)

CHASTE BERRY



- Chaste Tree suppresses the over-production of estrogen and is widely used to **balance hormones** and treat **premenstrual symptoms**
- Chaste Tree also reduces estrogen levels while elevating progesterone levels, and has strong anti-inflammatory, antibacterial and antifungal effects which helps in shrinking uterine fibroids

CHASTE BERRY

Vitex agnus-castus extracts for female reproductive disorders: a systematic review of clinical trials.

van Die MD, *Planta Med.* 2013 May;79(7):562-75. doi: 10.1055/s-0032-1327831.



Objective of this systematic review was to evaluate the evidence for the efficacy and safety of Vitex extracts from randomized, controlled trials investigating women's health

- In premenstrual dysphoric disorder, one study reported Vitex to be equivalent to fluoxetine, while in the other, fluoxetine outperformed Vitex.
- In latent hyperprolactinemia, one trial reported it to be superior to placebo for reducing TRH-stimulated prolactin secretion, normalising a shortened luteal phase, increasing mid-luteal progesterone and 17 β -oestradiol levels.

CHASTE BERRY

- While the other found Vitex comparable to bromocriptine for reducing serum prolactin levels and ameliorating cyclic mastalgia.
- Adverse events with Vitex were mild and generally infrequent.

The results from randomised, controlled trials to date suggest benefits for Vitex extracts in the treatment of:

- Premenstrual syndrome
- Premenstrual dysphoric disorder
- Latent hyperprolactinemia

The premenstrual syndrome, premenstrual mastodynia, fibrocystic mastopathy and infertility have often common roots: effects of extracts of chasteberry (*Vitex agnus castus*) as a solution



Dana Seidlova-Wuttke et al, (2017)

- The results from all trials suggested that VAC extracts ameliorated premenstrual symptoms including mastodynia, premenstrual dysphoric disorder and latent hyperprolactinemia. Cystic mastopathy and sterility due to corpus luteum insufficiencies were also beneficially influenced.
- Adverse events with VAC were mild and generally infrequent.

The premenstrual syndrome, premenstrual mastodynia, fibrocystic mastopathy and infertility have often common roots: effects of extracts of chasteberry (Vitex agnus castus) as a solution



Many studies, some of which are randomised, controlled trials support the efficacy and tolerability of Vitex Agnus-Castus extracts (VAC or Chasteberry) in the treatment of:

- **PMS**
- Premenstrual mastodynia* and mastalgia*
- **Premenstrual dysphoric disorder ****
- Infertility due to Luteal Phase Defect
- Fibrocystic mastopathy

*Pain in the breast

**Similar to premenstrual syndrome (PMS) but is more serious

The premenstrual syndrome, premenstrual mastodynia, fibrocystic mastopathy and infertility have often common roots: effects of extracts of chasteberry (*Vitex agnus castus*) as a solution



Dana Seidlova-Wuttke et al, (2017)

- The main mechanism of action of Chasteberry, which is behind all the its benefits, is its **dopaminergic action**.
- VAC extracts may contain dopaminergic compounds which inhibit pituitary prolactin release  So acting on **decreasing Prolactin level** makes VAC an option in the treatment of the mentioned cases since these diseases have a common root which is hyper-prolactinemia.

The premenstrual syndrome, premenstrual mastodynia, fibrocystic mastopathy and infertility have often common roots: effects of extracts of chasteberry (Vitex agnus castus) as a solution



Dana Seidlova-Wuttke et al, (2017)

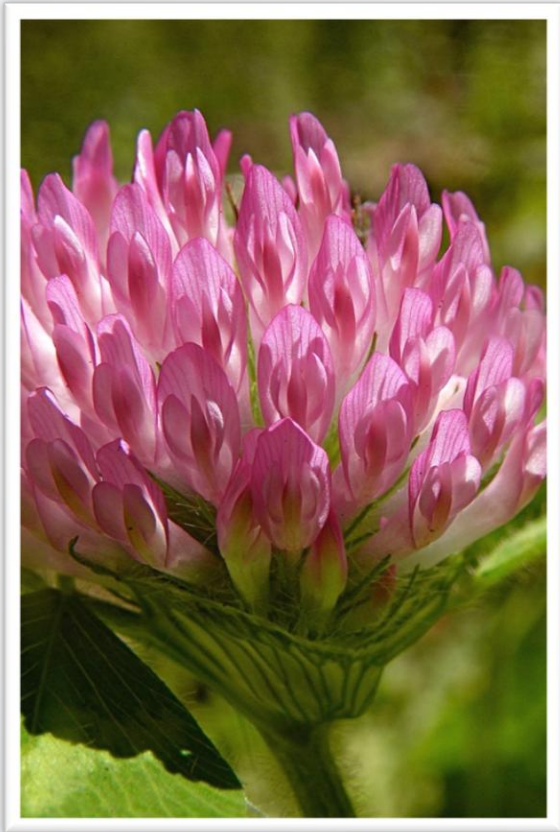
- Vitex could increase LH levels and decrease FSH levels thus it could help to normalize the ratio between Estrogen and Progesterone.
- *“Actions upon the pituitary gland-specifically on the production of a hormone called luteinizing hormone (LH). This indirectly increases [progesterone](#) production and helps regulate the menstrual cycle and helps in the cases of luteal phase defect.”*

MOTHERWORT

- Motherwort is used world-wide for a range of menstrual and reproductive problems
- It contains **Leonurine** which **relaxes the smooth muscles** of the uterus, **preventing cramping**, and **reducing menstrual pain**
- Motherwort regulates menstruation, reduces cramping and reduces heavy bleeding



RED CLOVER



- As a source of nutrients, Red Clover serves as a replenishment for the body after many months of heavy bleeding
- Red Clover contains tocopherol (Vitamin E), proteins, salicylates and coumarins, which **reduce blood clotting**
- Red Clover has been used for hundreds of years to **improve liver function**, which is essential to **hormonal balance**. Healthy estrogen metabolism must be reestablished to reverse estrogen dominance. Restoring and maintaining healthy **detoxification** can help in establishing hormonal balance.

RED CLOVER



- It is one of the best **blood purifying** herbs aiding in detoxification.
- Red clover aids in improved circulatory function, this is very important for many functions of the body, including the reproductive functions.
- Red Clover has been shown to increase cervical mucous, aiding in alleviating vaginal dryness, in addition to its important role in improving uterine disorders as Uterine Fibroids and Endometriosis.
- Red Clover has **Antispasmodic** and sedative properties, as a phytoestrogen (plant estrogen), it **helps balance the body's natural estrogen levels and regulate the menstrual cycle.**

RED CLOVER

RANDOMISED PLACEBO CONTROLLED PILOT STUDY ON COMPARING RED CLOVER (P-07) VERSUS PLACEBO FOR THE TREATMENT OF PREMENSTRUAL SYNDROME

Taher, S. et al, (2009).

- Red Clover (isoflavinoid phytoestrogens) has a similar structure to Estrogens with 100-1000 times weaker than Oestradiol.
- 19 patients randomly assigned to either placebo or active 40 mg Isoflavone treatment in a randomized double-blind protocol.
- The Primary outcome measure is Change in total, exponentially smoothed, average maximum score (total-ESAm_{max}) of 7 common premenstrual symptoms derived from Trigg's trend analysis from baseline (average of the 2 screening cycle) during the double-blinded treatment phase.

RED CLOVER

- Results: There is significant difference in total and median-ESAmax in Isoflavon (P-07) group before and after treatment (36.48% symptom reduction (P value <0.04). In placebo group, there is moderate difference in total and median-ESAmax before and after treatment (29.07% symptom reduction). In Isoflavon (P-07) group, there is evidence of a significant difference of crying, fatigue and swelling symptoms pre and post-treatment (P<0.05) however the evidence of difference was borderline in anxiety, and depression symptom (P=0.07).
- Conclusion: Administration of an isoflavone extracted from red clover was associated with a significant reduction in some premenstrual syndrome symptoms in total and in particular psychological symptoms.

BURDOCK ROOT



- Burdock Root is **another blood purifying herb**, used to support the liver and **detoxify excess estrogen and xenoestrogens**
- It **also has diuretic properties**, which help the body get rid of retained water, **reducing bloating and swelling**
- Its **anti-inflammatory properties** also assist in shrinking fibroids in size
- Burdock Root is rich in Arctigenin, which **inhibits the growth of tumors.**

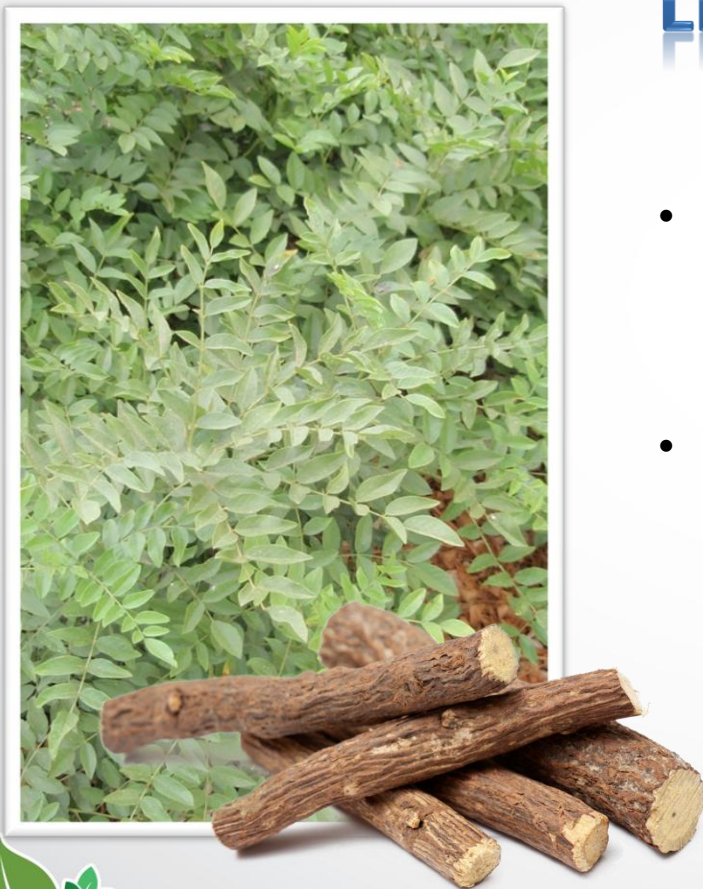
BURDOCK ROOT



- Burdock Root is rich in Arctigenin, which **inhibits the growth of tumors.**
- **Arctigenin (AR)** is a lignan found in certain plants of the Asteraceae, including the greater burdock and Saussurea heteromalla. It has shown **antiviral and anticancer effects in vitro.**
- AR exhibits potent anti-inflammatory activities by inhibiting inducible nitric oxide synthase (iNOS) via modulation of several cytokines. Due to its potent anti-inflammatory effects, AR may serve as a potential therapeutic compound against both acute inflammation and various chronic diseases. *Qiong GAO, et al (2018)*

LICORICE ROOT

- Licorice Root is used to treat **fluid retention**, helping with bloating and weight gain during menstruation
- Licorice Root is also used to calm the digestion, and **when combined with Red Clover, it supports the liver in breaking down excess estrogen**



GOLDENSEAL



- Goldenseal has an **anti inflammatory** effect
- It is used to maintain the health of mucus membranes - the tissue surrounding the internal genitals
- The **berberine** in Goldenseal is **antibacterial** and **antiviral**, which helps the body fight infection and strengthens the immune system, especially when combined with Echinacea.



ECHINACEA

- Echinacea has been used for over 400 years as a general cure-all. Recent studies suggest that Echinacea enhances the immune system, **reduces inflammation** and **relieves pain**.
- Echinacea/red root compound **modulates toxin elimination, immune system, and inflammatory mediators**. *Allan Warshowsky, MD, FACOG, ABIHM. Uterine Fibroids (Leiomyomata), Chapter 58.*



How to take?
As a dietary supplement, take 2 capsules per day or as directed by your health care provider for 3-6 months.

THANK
YOU

